Practice Problems
Calculus Physics II

**Name:** Key **Date:** _____

## Chapter 58 Refraction of Light

1) If the index of refraction of a material is 2, this means that light travels

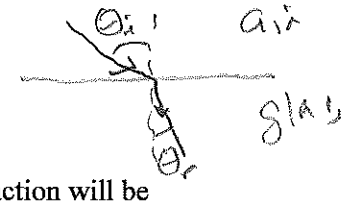
$n = \frac{c}{v}$ $\quad v = \frac{c}{2}$

A) 2 times faster in air as it does in vacuum.
B) 2 times faster in the material as it does in air.
C) 2 times faster in vacuum as it does in the material.
D) 2 times faster in the material than it does in vacuum.
E) 1/2 as fast in air as it does in the material.

2) In which material does light travel fastest?

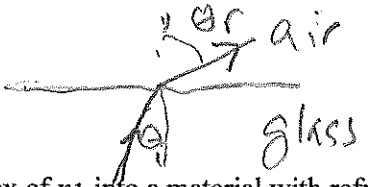
A) glass
B) water
C) diamond
D) air
E) same for all

3) Light enters glass from air. The angle of refraction will be




A) greater than the angle of incidence.
B) equal to the angle of incidence.
C) less than the angle of incidence.

4) Light enters air from water. The angle of refraction will be




A) greater than the angle of incidence.
B) equal to the angle of incidence.
C) less than the angle of incidence.

5) Light goes from material having a refractive index of $n_1$ into a material with refractive index $n_2$. If the refracted light is bent away from the normal, what can you conclude about the indices of refraction?

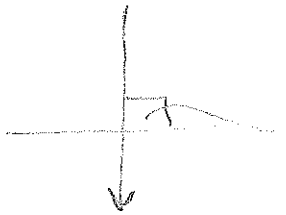
A) $n_1 > n_2$
B) $n_1 \geq n_2$
C) $n_1 < n_2$
D) $n_1 \leq n_2$
E) $n_1 = n_2$

6) When a piece of clear plastic with the same index of refraction as water is submerged in water, which statement is correct for a ray of light incident of the plastic from the water?

A) There is reflection but no refraction of the light.
B) There is both reflection and refraction of the light.
C) There is no reflection and no refraction of the light.
D) There is no reflection but there is refraction of the light.

7) A beam of light in air has an incident angle of 0° to the normal (90° to the surface) to a piece of glass, is the beam of light reflected or refracted?

A) It is both reflected and refracted.
B) It is only reflected not refracted.
C) It is not reflected but is refracted.
D) It is not reflected or refracted.




8) Mike dips a glass rod into vegetable oil. The submerged part of the rod can't be seen because

A) both the oil and glass have the same index of refraction.
B) light has different speeds in the oil and glass.
C) of internal reflection.
D) refraction is unbent before light gets to your eye.

9) The speed of light in a certain medium is $2.2 \times 10^8$ m/s. What is the index of refraction of this medium?

$n = \frac{c}{v} = \frac{3 \times 10^8 \text{ m/s}}{2.2 \times 10^8 \text{ m/s}} = 1.36$

10) The index of refraction of a diamond is 2.42. What is the speed of blue light with a wavelength of 480 nm (in vacuum), when the light is in the diamond?

$n = \frac{c}{v} \quad | \quad v = \frac{c}{n} = \frac{3 \times 10^8 \text{ m/s}}{2.42} = 1.24 \times 10^8 \text{ m/s}$

11) The index of refraction of a diamond is 2.42. What is the wavelength of the 480 nm (in vacuum) light in the diamond?

$\lambda_{material} = \frac{\lambda_{vacuum}}{n} = \frac{480 \text{ nm}}{2.42} = 198.3 \text{ nm} = 198.3 \times 10^{-9} \text{ m}$

12) A red laser pointer ($\lambda = 630$ nm) goes from air ($n_{air} = 1.0$) to glass ($n_{glass} = 1.6$), what is the speed of the light in the glass?

$n = \frac{c}{v} \quad | \quad v = \frac{c}{n} = 1.875 \times 10^8 \text{ m/s}$

13) A red laser pointer ($\lambda = 630$ nm) goes from air ($n_{air} = 1.0$) to glass ($n_{glass} = 1.6$), what is the wavelength of the light in the glass?

$\lambda_{material} = \frac{\lambda_{vac}}{n} = 394 \text{ nm} = 394 \times 10^{-9} \text{ m}$

14) A red laser pointer ($\lambda = 630$ nm) goes from air ($n_{air} = 1.0$) to glass ($n_{glass} = 1.6$), what is the frequency of the light in the glass?

$v = c = f\lambda \quad | \quad f = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{630 \times 10^{-9} \text{ m}} = 4.76 \times 10^{14} \text{ Hz}$

For air/vacuum

15) A beam of light in air enters a glass slab with an index of refraction of 1.40 at an angle of incidence of 30.0°. What is the angle of refraction?

$n_{air} \sin\theta_{air} = n_g \sin\theta_g$

Solve for $\theta_g$

$\theta_g = \sin^{-1}\left(\frac{n_{air}\sin(\theta_{air})}{n_g}\right)$

$\theta_g = 20.9°$

16) A beam of light in water ($n = 1.33$) enters a glass slab ($n = 1.50$) at an angle of incidence of 60.0°. What is the angle of refraction in the glass?

$n_w \sin\theta_w = n_g \sin\theta_g$

Solve for $\theta_g$

$\theta_g = \sin^{-1}\left(\frac{n_w \sin\theta_w}{n_g}\right) = 50.2°$

17) A flashlight beam makes an angle of 60° with the normal to the surface of a calm lake (in air) before it enters the water. What is the refracted angle of the beam in the water? The index of refraction of water is 1.33.

$n_a \sin\theta_a = n_w \sin\theta_w$

Solve for $\theta_w$

$\theta_w = \sin^{-1}\left(\frac{n_a \sin\theta_a}{n_w}\right) = 40.6°$

18) A scuba diver has a flashlight under water (n = 1.33) and shines a beam of light at an angle of 20° to the normal relative to the water/air interface. If we assume the interface is smooth (no waves) at what angle relative to the normal does the light exit?

$n_w \sin\theta_w = n_a \sin\theta_a$

Solve for $\theta_a$

$\theta_a = \sin^{-1}\left(\frac{n_w \sin\theta_w}{n_a}\right) = 27°$

19) A beam of light traveling in air strikes a slab of transparent material. The incident beam makes an angle of 40° with the normal, and the refracted beam make an angle of 26° with the normal. What is the speed of light in the transparent material? ($c = 3.0 \times 10^8$ m/s)

need $n_{slab}$

$n_a \sin\theta_a = n_s \sin\theta_s$

$n_s = \frac{n_a \sin\theta_a}{\sin\theta_s} = 1.47$

$n = \frac{c}{v}$

$v = \frac{c}{n} = 2.04 \times 10^8$ m/s

20) A glass plate with an index of refraction is 1.57 is immersed in an oil having a refractive index of 1.40. The beam of light is incident on the glass plate from the oil at an angle of $\Theta_i$, if the refracted angle in the glass is 25°, what is the incident angle the light entering the glass?

$n_o \sin\theta_o = n_g \sin\theta_g$

Solve For $\theta_o$

$\theta_{o,i} = \sin^{-1}\left(\frac{n_g \sin\theta_t}{n_o}\right) = 28.3°$

21) Three different materials are stacked on top of each other, given the information below, determine the refracted angle $\Theta_3$ in material # 3.

$n_1 = 1.65$ and $\Theta_1 = 20°$
$n_2 = 1.5$
$n_3 = 1.3$

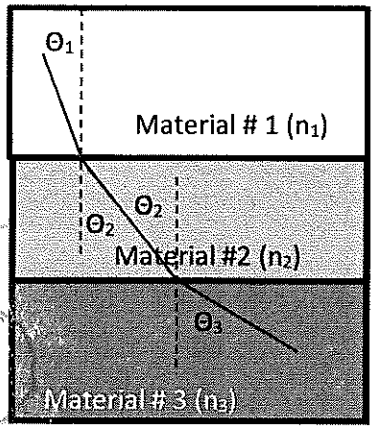
$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$\theta_2 = \sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right)$

$\theta_2 = 22.1°$

$n_2 \sin\theta_2 = n_3 \sin\theta_3$

$\theta_3 = \sin^{-1}\left(\frac{n_2 \sin\theta_2}{n_3}\right) = 25.7°$




22) What percentage of the light is reflected from the interface of glass (n = 1.6) and water (1.33) if the light is at 90° to the surface (0° to the normal to the surface)?

$R = \left[\frac{n_1 - n_2}{n_1 + n_2}\right]^2 = 0.008$

$\%R = R(100\%) = 0.8\%$

23) What is the difference in refracted angle for a red beam of light ($n_{red} = 1.56$) is incident at the same angle $\Theta_i = 30°$ from air, as a blue beam of light ($n_{blue} = 1.59$)?

$\theta_{red} = \sin^{-1}\left(\frac{n_a \sin\theta_a}{n_r}\right) = 18.7°$

$\theta_{blue} = \sin^{-1}\left(\frac{n_a \sin\theta_a}{n_b}\right) = 18.3°$

$\theta_{red} - \theta_{blue} = 0.4°$

24) A scuba diver has a flashlight under water (n = 1.33) and shines a beam of light at an angle of 60° to the normal relative to the water/air interface. If we assume the interface is smooth (no waves) at what angle relative to the normal does the light exit?

Total internal reflection

$\theta_c = \sin^{-1}\left(\frac{n_L}{n_H}\right) = 48.7°$

60° is passed 48.7°, no light transmitted

25) What is the critical angle for total internal reflection for light traveling from glass (n = 1.6) to water (n = 1.33)?

$\theta_c = \sin^{-1}\left(\frac{n_L}{n_H}\right) = \sin\left(\frac{1.33}{1.6}\right) = 56.2°$

26) If the critical angle of a glass to air interface is 45°, what is the index of refraction of the glass?

$\sin\theta_c = \frac{n_L}{n_H} = \frac{n_{air}}{n_g}$



$n_g = \frac{n_{air}}{\sin\theta_c} = 1.41$

27) A mirage is a result of atmospheric

A) reflection.
B) dispersion.
C) scattering.
D) aberrations.
E) refraction.

28) An inferior mirage bends light due to

A) Hot air having a larger index of refraction than cold air
B) Hot air having a smaller index of refraction than cold air

29) The twinkling of the stars is a result of atmospheric

A) reflection.
B) refraction.
C) scattering.
D) dispersion.
E) aberrations.

30) Refraction causes the bottom of a swimming pool to appear

A) farther down than it actually is.
B) closer to the surface than it actually is.
C) neither of these

31) When spearing a fish in a clear lake with a regular spear, to compensate for refraction, throw your spear

A) above the sighted fish.
B) directly at the sighted fish.
C) below the sighted fish.

32) A penny is at the bottom of a fountain, if the penny appears to be 0.5 meters deep, what is the actual depth of the water (n = 1.33) in the fountain?

$d' = d\left[\frac{n_L}{n_H}\right]$

Solve For d

$d = \frac{d'}{\left(\frac{n_L}{n_H}\right)} = 0.665\ m$

KEY

1) C
2) D
3) C
4) A
5) A
6) C
7) B
8) A
9) 1.36
10) $1.24 \times 10^8$ m/s
11) 198.3 nm
12) $1.875 \times 10^8$ m/s
13) 394 nm
14) $4.76 \times 10^{14}$ Hz
15) 20.9°
16) 50.2°
17) 40.6°
18) 27°
19) $2.04 \times 10^8$ m/s
20) 28.3°
21) 25.7°
22) 0.8 %
23) 18.7° - 18.3° = 0.4°
24) No light exits, the beam is past the critical angle.
25) 56.2°
26) 1.41
27) E
28) B
29) B
30) B
31) C
32) 0.665 m